

Paul Zinn-Justin

Professor, School of Mathematics and Statistics, The University of Melbourne

CURRICULUM VITAE

pzinn@unimelb.edu.au | <http://blogs.unimelb.edu.au/paul-zinn-justin/>

School of Mathematics and Statistics, The University of Melbourne, VIC 3010, Australia

[ORCID](#) | [zbMATH](#) | [Google Scholar](#)

PROFESSIONAL EXPERIENCE

2020–	Professor at the School of Mathematics and Statistics of the University of Melbourne (Australia).
2016–2020	ARC Future Fellow at the School of Mathematics and Statistics of the University of Melbourne (Australia).
2013–2016	Directeur de recherches (senior scientist) CNRS at LPTHE (UMR 7589), Université Pierre et Marie Curie (France).
2012	“Research Member” of the MSRI program “Random Spatial Processes” (Berkeley).
2009–2010	“Visiting Professor” of the KNAW (Royal Netherlands Academy of Arts and Sciences), visiting the University of Amsterdam.
2008–2013	Chargé de recherches CNRS at LPTHE (UMR 7589), Université Pierre et Marie Curie (France).
2005–2008	Chargé de recherches CNRS at LPTMS (UMR 8626), Université Paris-Sud (France).
2004–2005	Chargé de recherches CNRS at Laboratoire Poncelet (LIFR-MIIP, UMI 2615), Moscow (Russia).
2000–2004	Chargé de recherches (junior scientist) CNRS at LPTMS (UMR 8626), Université Paris-Sud (France).
2000	Visitor at C.N. Yang Institute for Theoretical Physics, SUNY Stony Brook.
1999–2000	Post-doctoral fellow at the New High Energy Theory Center, Rutgers University.
1998–1999	National service.
1995–1998	PhD in theoretical physics at the Laboratoire de Physique Théorique de l’Ecole Normale Supérieure de Paris, supervisor V. Kazakov.

RESEARCH INTERESTS

- Algebraic combinatorics, symmetric functions; Schubert calculus, enumerative geometry.
- Exactly solvable models in statistical mechanics, quantum integrability, Yang–Baxter equation, and quantum groups.
- Random matrices, and connections between mathematical physics and enumerative combinatorics.
- Computational and experimental mathematics, including algorithm design, computer algebra and machine learning.

PRIZES, AWARDS

2016–2020	Australian Research Council Future Fellowship “From quantum integrable systems to algebraic geometry and combinatorics”.
2011–2016	European Research Council grant “Loop models, Integrability and Combinatorics”.
2010–2014	Prime d’excellence scientifique (scientific excellency award) from CNRS.
2009	Visiting Professorship award from the Royal Netherlands Academy of Arts and Sciences.

TEACHING EXPERIENCE

2019–2026	<i>Discrete Mathematics, Linear Algebra, Group theory and linear algebra, Enumerative combinatorics, Exactly solvable models, Experimental mathematics</i> (48h/36h) courses at the University of Melbourne.
2022	LAWRGe lectures at the University of Southern California: <i>Schubert Calculus and Quantum Integrability</i> (14h).
2019	Mini-course at the University of Virginia: <i>Quantum integrability and symmetric polynomials</i> (10h).
2019	Mini-course at the University of Melbourne: <i>Schubert calculus</i> (12h).
2018	Mini-course at the University of Melbourne: <i>Geometry, Integrability and symmetric functions</i> (6h).
2015	Lectures at the Department of Mathematics of the Higher School of Economics (Moscow): <i>Geometry, Integrability and symmetric functions</i> (8h).
2013–2015	<i>Intégrabilité quantique</i> , Masters course at Université Pierre et Marie Curie (30h).
2012	Chern–Simons lectures at the University of California at Berkeley: <i>Exactly solvable tiling models, Schur functions and the Littlewood–Richardson rule</i> (6h).
2008	Course at Les Houches summer school: <i>Integrability and combinatorics: selected topics</i> .
2007	Lectures at Barcelona University: <i>Quantum integrability and Razumov–Stroganov type conjectures</i> .
1997–1998	Teaching Assistant at ENS Paris: mathematics course.
1995–1997	Oral examiner at preparatory school Louis-Le-Grand (bachelor level).

STUDIES AND ACADEMIC HONOURS

2008	Habilitation à Diriger des Recherches “Six-Vertex, Loop and Tiling models: Integrability and Combinatorics” (Université Pierre et Marie Curie, Paris 6).
1998	PhD “Quelques applications de l’Ansatz de Bethe”, mention très honorable avec les félicitations du jury (highest mark) (Université Pierre et Marie Curie, Paris 6).
1996	Mathematics “Agrégation” (teaching diploma).
1994–1995	D.E.A. (Master’s) in theoretical physics, ENS.
1993–1994	Licence et maîtrise in mathematics, licence in physics (Bachelor’s degrees).

- 1993** Admission at Ecole Normale Supérieure (ENS, Paris) on the Mathematics–Physics–Computer Science entrance exam.
- 1992** Silver medal at the International Mathematical Olympiads.

ADMINISTRATION OF RESEARCH

- Organisation of scientific meetings:
 - conference “Formal Power Series and Algebraic Combinatorics 2026”, Seattle (Program committee member).
 - conference “At the crossroads of physics and mathematics: the joy of integrable combinatorics — a conference in honour of Philippe Di Francesco’s 60th birthday”, 24–26 June 2024, Saclay (France).
 - conference “Integrability, Combinatorics and Representations ’19”, 2–6 September 2019, Giens (France).
 - workshop “Geometric R -matrices”, 18–22 December 2017, MATRIX (Australia).
 - program “New trends in integrable models”, 15 August–25 November 2016, International Institute of Physics (Natal, Brazil).
 - program “Statistical Mechanics, Integrability and Combinatorics”, 11 May–3 July 2015, and conference “Lattice Models: Exact Methods and Combinatorics”, 18–22 May 2015, Galileo Galilei Institute (Florence, Italy).
 - conference “Integrability and Combinatorics ’14”, Giens (France), 23–27 June 2014.
 - conference “Formal Power Series and Algebraic Combinatorics 2013”, Paris (Program committee member).
 - conference “Random Matrices and Integrable Systems”, Les Houches, 5–9 March 2012.
 - conference “GranMa ’11”, IHP (Paris), 3–6 October 2011.
 - conference “Geometry and Integrability in Mathematical Physics ’08”, CIRM Luminy (France), 15–19 September 2008.
 - conference “Geometry and Integrability in Mathematical Physics ’06”, Moscow (Russia), 15–19 June 2006.
 - workshop “Combinatorial methods in Physics and Knot Theory”, Moscow (Russia), 21–25 February 2005.
- Supervision of six PhD students (T. Fonseca 2007–2010, A. Garbali 2012–2015, A. Gunna 2020–2024, A. Stoyan 2023–2026, Junru Hu 2026–, Jakob Sloan 2026–), of ten post-docs (L. Cantini 2006–2008, K. Kalle 2007–2008, M. Wheeler and D. Betea 2012–2014, A. Ponsaing 2013–2014, P. Finch 2014–2015, I. Halacheva 2018–2020, J. Lamers 2020–2021, J.-E. Bourgine 2022–2023, L. Cassia 2023–2025). L. Cantini later received the CNRS Bronze Medal. Twelve of these are still in academia, among whom five have permanent positions; the remaining four successfully transitioned to the private sector. Supervision of seven Master students and three Bachelor students summer internships.
- Recipient of four Discovery Projects “Multivariate polynomials: combinatorics and applications” (2014–2017), “Matrix product multi-variable polynomials from quantum algebras” (2019–2023), “Elliptic Schubert calculus” (2021–2026), “Shuffle algebras and vertex models” (2024–2026), and a Future Fellowship “From quantum integrable systems to algebraic geometry and combinatorics” (2016–2020) from the ARC.
- Recipient of an ERC consolidator grant for the project “Loop models, Integrability and Combinatorics”, 2011–2016 (budget: 900k€).
- “Scientist in charge” of the Marie Curie Research Training Network “ENRAGE” (Random Geometry and Random Matrices: From Quantum Gravity to Econophysics), 2005–2009.

- Coordinator of the ANR program “GIMP” (Geometry and Integrability in Mathematical Physics), 2006–2009.
- Former editor for *Communications in Mathematical Physics*; editor for *Algebraic Combinatorics*.
- Referee for numerous journals (Commun. Math. Phys., Lett. Math. Phys., Electronic J. of Combinatorics, Adv. Appl. Math., J. of Combin. Theory A, JSTAT, J. of Physics A, etc); expert for several research organisations (ANR – France, NWO – Netherlands, ESF – Europe, ARC – Australia).
- Member of the Committee of Academic Sponsors of SLMATH (Berkeley). Former member of the Diversity and Inclusion committee (Melbourne University). Former member of the laboratory council (LPTMS and LPTHE).
- PhD referee (A. Ponsaing, L. Poulain d’Andecy).
- Member of the European Marie Curie Research Training Network “ENIGMA”, 2005–2008, of the ESF program “MISGAM”, 2005–2009, and of the ANR program “GranMa”, 2009–2011.

SUMMARY OF PUBLICATIONS

- 80 publications in major international peer-reviewed journals.
- 7 chapters of books.
- 8 preprints.
- 1 popular science article.
- “Habilitation à Diriger des Recherches” thesis was published as a monograph.
- h-index: 33 (Google Scholar, April 2026)

SUMMARY OF CONFERENCES

- Organised conferences: 12
- Invited oral presentations in international conferences and schools: 98
- Other oral presentations: 67

SELECTED INVITED AND PLENARY CONFERENCE TALKS

- Conference “FPSAC”, UC Davis (USA), 17–21 July 2023: *Schubert puzzles as exactly solvable models* (plenary talk).
- International Congress of Mathematical Physics, Montreal (Canada), 23–28 July 2018: *From quantum integrability to Schubert calculus*.
- Conference “Integrability and Algebraic Combinatorics”, IPAM UCLA (USA), 15–19 April 2024: *The MacNeille completion of the type B Bruhat poset*.
- Conference “Quantum integrability and quantum Schubert calculus”, Kavli Royal Society Centre (UK), 11–13 June 2018: *Schubert puzzles and quantum integrability*.
- Symposium FoPM Tokyo (Japan), 6–8 February 2023: *From exactly solvable models of statistical mechanics to combinatorics*.
- Main speaker at the 76th “Séminaire Lotharingien de Combinatoire”, 3–6 April 2016: *Razumov–Stroganov Correspondences and the Geometry of Schubert Varieties*.

- Professional expertise in algorithm design.
- High level programming skills in:
 - Traditional programming languages: python, c, c++, C#, Lisp, assembly language. . . ;
 - Web-based languages and related: HTML/CSS/JavaScript/TypeScript, PHP/SQL;
 - Mathematics-related languages: Macaulay2, Mathematica.
- Contributes to various open source projects.
Developer for (since 2025, “author of”) the open source software “Macaulay2”:
 - Bug fixes, implementation of new functions.
 - Rewrite of the output engine.
 - Macaulay2Web, a web-based interface using the $\text{K}\text{A}\text{T}\text{E}\text{X}$ technology.
- Contributes entries to the On-Line Encyclopedia of Integer Sequences.
- Various projects can be found on my website and on my GitHub page.

PUBLICATIONS

- [95] Paul Zinn-Justin. q -difference equations for two classes of pattern avoiding permutations, 2026. [arXiv:2608.09078](#).
- [94] Geordie Williamson, Oded Yacobi, and Paul Zinn-Justin. Generating Hadamard matrices with transformers, 2026. [arXiv:2604.11101](#).
- [93] Ajeeth Gunna, Michael Wheeler, and Paul Zinn-Justin. Inhomogeneous q -Whittaker polynomials I: Duality and expansions, 2025. [arXiv:2512.04468](#).
- [92] Bruce W. Westbury and Paul Zinn-Justin. A uniform R -matrix for adjoint representations. *Bulg. J. Phys.*, 52(1):159–164, 2025.
- [91] Allen Knutson and Paul Zinn-Justin. Hybrid pipe dreams for the lower-upper scheme, 2025. [arXiv:2509.01857](#).
- [90] Ajeeth Gunna, Michael Wheeler, and Paul Zinn-Justin. Structure constants for spin Hall–Littlewood functions, 2025. [arXiv:2504.19205](#).
- [89] Allen Knutson and Paul Zinn-Justin. Generic pipe dreams, lower-upper varieties, and Schwartz–MacPherson classes. In *Proceedings of the 37th Conference on Formal Power Series and Algebraic Combinatorics (San Diego, 2006)*, volume 93B, 2025. [arXiv:2411.11208](#).
- [88] Bruce W. Westbury and Paul Zinn-Justin. A uniform trigonometric R -matrix for the exceptional series, 2024. [arXiv:2406.01348](#).
- [87] Paul Zinn-Justin. Integrability and combinatorics. In *Encyclopedia of Mathematical Physics*, volume 3. 2025. chapter of the Encyclopedia of Mathematical Physics. [arXiv:2404.13221](#).
- [86] Yaping Yang and Paul Zinn-Justin. Higher spin representations of the Yangian of $\mathfrak{sl}_2$ and R -matrices, 2024. [arXiv:2403.17433](#).
- [85] Allen Knutson and Paul Zinn-Justin. Schubert puzzles and integrability III: separated descents, 2023. [arXiv:2306.13855](#).
- [84] Paul Zinn-Justin. The *CotangentSchubert* Macaulay2 package. *Journal of Software for Algebra and Geometry*, 14:73–85, 2024. <https://www.unimelb-macaulay2.cloud.edu.au/#tutorial-CotangentSchubert>.
- [83] Alexandr Garbali and Paul Zinn-Justin. Shuffle algebras, lattice paths and the commuting scheme. In *Hypergeometry, Integrability and Lie Theory*. 2022. Editors Erik Koelink, Stefan Kolb, Nicolai Reshetikhin and Bart Vlaar. [arXiv:2110.07155](#).

- [82] Allen Knutson and Paul Zinn-Justin. Schubert puzzles and integrability II: multiplying motivic Segre classes. *Comm. A.M.S.*, 6:68–152, 2026. [arXiv:2102.00563](#).
- [81] Ajeeth Gunna and Paul Zinn-Justin. Vertex models for canonical Grothendieck polynomials and their duals. *Algebraic Combinatorics*, 6(1):109–163, 2023. [arXiv:2009.13172](#).
- [80] Andrew Elvey Price and Paul Zinn-Justin. The six-vertex model on random planar maps revisited. *Journal of Combinatorial Theory, Series A*, 196:105739, 2023. [arXiv:2007.07928](#).
- [79] Paul Zinn-Justin. The trigonometric E_8 R -matrix. *Lett. Math. Phys.*, 110:3279–3305, 2020. [arXiv:2004.02044](#).
- [78] Paul Zinn-Justin. Honeycombs for Hall polynomials. *Electron. J. Combin.*, 27:P2.23, 2020. [arXiv:1909.10720](#).
- [77] Dmitri Bykov and Paul Zinn-Justin. Higher spin $\mathfrak{sl}_2$ R -matrix from equivariant (co)homology. *Lett. Math. Phys.*, 110:2435–2470, 2020. [arXiv:1904.11107](#).
- [76] Mireille Bousquet-Mélou, Andrew Elvey Price, and Paul Zinn-Justin. Eulerian orientations and the six-vertex model on planar maps. In *Proceedings of the 31st Conference on Formal Power Series and Algebraic Combinatorics (Ljubljana, 2019)*, volume 82B, 2020. <http://fpsac2019.fmf.uni-lj.si/resources/Proceedings/129.pdf>, [arXiv:1902.07369](#).
- [75] Iva Halacheva, Allen Knutson, and Paul Zinn-Justin. Restricting Schubert classes to symplectic Grassmannians using self-dual puzzles. In *Proceedings of the 31st Conference on Formal Power Series and Algebraic Combinatorics (Ljubljana, 2019)*, volume 82B, page Art. 83, 2020. <http://fpsac2019.fmf.uni-lj.si/resources/Proceedings/154.pdf>, [arXiv:1811.07581](#).
- [74] Andrew Conway, Anthony Guttman, and Paul Zinn-Justin. 1324-avoiding permutations revisited. *Advances in Applied Mathematics*, 96:312–333, 2018. [arXiv:1709.01248](#).
- [73] Allen Knutson and Paul Zinn-Justin. Schubert puzzles and integrability I: invariant trilinear forms. *Comm. A.M.S.*, 6:1–67, 2026. [arXiv:1706.10019](#).
- [72] Paul Zinn-Justin. Loop models and K -theory. *SIGMA Symmetry Integrability Geom. Methods Appl.*, 14:paper 69, 48, 2018. [arXiv:1612.05361](#).
- [71] Allen Knutson and Paul Zinn-Justin. Grassmann–Grassmann conormal varieties, integrability, and plane partitions. *Annales de l’Institut Fourier*, 69(3):1087–1145, 2019. [arXiv:1612.04465](#).
- [70] Michael Wheeler and Paul Zinn-Justin. Littlewood–Richardson coefficients for Grothendieck polynomials from integrability. *J. Reine Angew. Math.*, 757:159–195, 2019. [arXiv:1607.02396](#).
- [69] Michael Wheeler and Paul Zinn-Justin. Hall polynomials, inverse Kostka polynomials and puzzles. *Journal of Combinatorial Theory, Series A*, 159:107–163, 2018. [arXiv:1603.01815](#).
- [68] Peter Finch, Robert Weston, and Paul Zinn-Justin. Theta function solutions of the quantum Knizhnik–Zamolodchikov–Bernard equation for a face model. *J. Phys. A*, 49(6), 2016. [arXiv:1509.04594](#).
- [67] Michael Wheeler and Paul Zinn-Justin. Refined Cauchy/Littlewood identities and six-vertex model partition functions: III. deformed bosons. *Adv. Math.*, 299:543–600, 2016. [arXiv:1508.02236](#).
- [66] Paul Zinn-Justin. Quiver varieties and the quantum Knizhnik–Zamolodchikov equation. *Theoretical and Mathematical Physics*, 185(3):1741–1758, 2015. [arXiv:1502.01093](#).
- [65] Peter Forrester, L. Dang-Zheng, and Paul Zinn-Justin. Equilibrium problems for Raney densities. *Nonlinearity*, 28(7):2265, 2015. [arXiv:1411.4091](#).
- [64] Anita Ponsaing and Paul Zinn-Justin. Type $\hat{C}$ Brauer loop schemes and loop model with boundaries. *Ann. Henri Poincaré*, 3(2):163–255, 2016. [arXiv:1410.0262](#).
- [63] D. Betea, Michael Wheeler, and Paul Zinn-Justin. Refined Cauchy/Littlewood identities and six-vertex model partition functions: II. proofs and new conjectures. *Journal of Algebraic Combinatorics*, pages 1–49, 2015. [arXiv:1405.7035](#).
- [62] Yacine Ikhlef, Robert Weston, Michael Wheeler, and Paul Zinn-Justin. Discrete holomorphicity and quantized affine algebras. *J. Phys. A*, 46(26):265205, 2013. [arXiv:1302.4649](#).

- [61] Tiago Fonseca and Paul Zinn-Justin. Higher spin polynomial solutions of quantum Knizhnik–Zamolodchikov equation. *Comm. Math. Phys.*, 328(3):1079–1115, 2014. [arXiv:1212.4672](#).
- [60] Roger Behrend, Philippe Di Francesco, and Paul Zinn-Justin. A doubly refined enumeration of alternating sign matrices and descending plane partitions. *J. Combin. Theory Ser. A*, 120(2):409–432, 2013. [arXiv:1202.1520](#).
- [59] Paul Zinn-Justin. Sum rule for the eight-vertex model on its combinatorial line. In *Symmetries, Integrable systems and representations*, volume 40, 2013. Eds Iohara, Morier-Genoud, Rémy. [arXiv:1202.4420](#).
- [58] R. Rimányi, V. Tarasov, A. Varchenko, and Paul Zinn-Justin. Extended Joseph polynomials, quantized conformal blocks, and a q -Selberg type integral. *J. Geom. Phys.*, 62(11):2188–2207, 2012. [arXiv:1110.2187](#).
- [57] Roger Behrend, Philippe Di Francesco, and Paul Zinn-Justin. On the weighted enumeration of alternating sign matrices and descending plane partitions. *J. Combin. Theory Ser. A*, 119(2):331–363, 2012. [arXiv:1103.1176](#).
- [56] Alice Guionnet, Vaughn Jones, Dimitri Shlyakhtenko, and Paul Zinn-Justin. Loop models, random matrices and planar algebras. *Comm. Math. Phys.*, 316(1):45–97, 2012. [arXiv:1012.0619](#).
- [55] Filippo Colomo, Andrei Pronko, and Paul Zinn-Justin. The arctic curve of the domain-wall six-vertex model in its anti-ferroelectric regime. *J. Stat. Mech. Theory Exp.*, page L03002, 2010. [arXiv:1001.2189](#).
- [54] K. Shigechi and Paul Zinn-Justin. Path representation of maximal parabolic Kazhdan–Lusztig polynomials. *J. Pure Appl. Algebra*, 216(11):2533–2548, 2012. [arXiv:1001.1080](#).
- [53] Paul Zinn-Justin. Jucys–Murphy elements and Weingarten matrices. *Lett. Math. Phys.*, 91(2):119–127, 2010. [arXiv:0907.2719](#).
- [52] Paul Zinn-Justin and Jean-Bernard Zuber. Knot theory and matrix integrals. In *The Oxford Handbook of Random Matrix Theory*. 2011. Eds Akemann, Baik and Di Francesco. [arXiv:1006.1812](#).
- [51] Paul Zinn-Justin. A conjectured formula for Fully Packed Loop configurations in a triangle. *Electron. J. Combin.*, 17:Research Paper 107, 2010. [arXiv:0911.4617](#).
- [50] Teo Banica, Benoit Collins, and Paul Zinn-Justin. Spectral analysis of the free orthogonal matrix. *Int. Math. Res. Not.*, 17:3286–3309, 2009. [arXiv:0901.1918](#).
- [49] Tiago Fonseca and Paul Zinn-Justin. On some ground state components of the $O(1)$ loop model. *J. Stat. Mech. Theory Exp.*, page P03025, 2009. [arXiv:0901.1679](#).
- [48] Paul Zinn-Justin. Littlewood–Richardson coefficients and integrable tilings. *Electron. J. Combin.*, 16:Research Paper 12, 2009. [arXiv:0809.2392](#).
- [47] Paul Zinn-Justin. *Six-vertex, loop and tiling models: integrability and combinatorics*. Lambert Academic Publishing, 2009. Habilitation thesis. <http://www.lpthe.jussieu.fr/~pzinn/publi/hdr.pdf>.
- [46] Paul Zinn-Justin. Integrability and combinatorics: selected topics. In *Exact Methods in Low-dimensional Statistical Physics and Quantum Computing*, volume 89. Oxford University Press, 2010. Eds Jacobsen, Ouvry, Pasquier, Serban, and Cugliandolo. <http://www.lpthe.jussieu.fr/~pzinn/semi/intcomb.pdf>.
- [45] Tiago Fonseca and Paul Zinn-Justin. On the doubly refined enumeration of Alternating Sign Matrices and Totally Symmetric Self-Complementary Plane Partitions. *Electron. J. Combin.*, 15:Research Paper 81, 35 pp, 2008. [arXiv:0803.1595](#).
- [44] Jan de Gier, Pavel Pyatov, and Paul Zinn-Justin. Punctured plane partitions and the q -deformed Knizhnik–Zamolodchikov and Hirota equations. *J. Combin. Theory Ser. A*, 116(4):772–794, 2009. [arXiv:0712.3584](#).
- [43] Philippe Di Francesco and Paul Zinn-Justin. Quantum Knizhnik–Zamolodchikov equation: reflecting boundary conditions and combinatorics. *J. Stat. Mech. Theory Exp.*, (12):P12009, 30 pp, 2007. [arXiv:0709.3410](#).
- [42] Alexander Razumov, Yuri Stroganov, and Paul Zinn-Justin. Polynomial solutions of q KZ equation and ground state of XXZ spin chain at $\Delta = -1/2$. *J. Phys. A*, 40(39):11827–11847, 2007. [arXiv:0704.3542](#).
- [41] Philippe Di Francesco and Paul Zinn-Justin. Quantum Knizhnik–Zamolodchikov equation, Totally Symmetric Self-Complementary Plane Partitions and Alternating Sign Matrices. *Theor. Math. Phys.*, 154(3):331–348, 2008. [arXiv:math-ph/0703015](#).

- [40] Paul Zinn-Justin. Loop model with mixed boundary conditions, q KZ equation and alternating sign matrices. *J. Stat. Mech. Theory Exp.*, (1):P01007, 16 pp, 2007. [arXiv:math-ph/0610067](#).
- [39] Allen Knutson and Paul Zinn-Justin. The Brauer loop scheme and orbital varieties. *J. Geom. Phys.*, 78:80–110, 2014. [arXiv:1001.3335](#).
- [38] Paul Zinn-Justin. Proof of the Razumov–Stroganov conjecture for some infinite families of link patterns. *Electron. J. Combin.*, 13(1):Research Paper 110, 15 pp, 2006. [arXiv:math/0607183](#).
- [37] Paul Zinn-Justin. Combinatorial point for fused loop models. *Comm. Math. Phys.*, 272(3):661–682, 2007. [arXiv:math-ph/0603018](#).
- [36] Philippe Di Francesco, Paul Zinn-Justin, and Jean-Bernard Zuber. Sum rules for the ground states of the $O(1)$ loop model on a cylinder and the XXZ spin chain. *J. Stat. Mech.*, page P08011, 2006. [arXiv:math-ph/0603009](#).
- [35] Philippe Di Francesco and Paul Zinn-Justin. From orbital varieties to alternating sign matrices. In *Proceedings of the 18th Conference on Formal Power Series and Algebraic Combinatorics (San Diego, 2006)*, 2006. <http://www-igm.univ-mlv.fr/~fpsac/FPSAC06/SITE06/papers/60.pdf>, [arXiv:math-ph/0512047](#).
- [34] Philippe Di Francesco and Paul Zinn-Justin. Quantum Knizhnik–Zamolodchikov equation, generalized Razumov–Stroganov sum rules and extended Joseph polynomials. *J. Phys. A*, 38(48):L815–L822, 2005. [arXiv:math-ph/0508059](#).
- [33] Allen Knutson and Paul Zinn-Justin. A scheme related to the Brauer loop model. *Adv. Math.*, 214(1):40–77, 2007. [arXiv:math.AG/0503224](#).
- [32] Philippe Di Francesco and Paul Zinn-Justin. Inhomogeneous model of crossing loops and multidegrees of some algebraic varieties. *Comm. Math. Phys.*, 262(2):459–487, 2006. [arXiv:math-ph/0412031](#).
- [31] Philippe Di Francesco and Paul Zinn-Justin. Around the Razumov–Stroganov conjecture: proof of a multi-parameter sum rule. *Electron. J. Combin.*, 12:Research Paper 6, 27 pp, 2005. http://www.combinatorics.org/Volume_12/Abstracts/v12i1r6.html, [arXiv:math-ph/0410061](#).
- [30] Philippe Di Francesco, Paul Zinn-Justin, and Jean-Bernard Zuber. Determinant formulae for some tiling problems and application to fully packed loops. *Ann. Inst. Fourier (Grenoble)*, 55(6):2025–2050, 2005. [arXiv:math-ph/0410002](#).
- [29] Paul Zinn-Justin. Conjectures on the enumeration of alternating links. In *Physical and numerical models in knot theory*, volume 36 of *Ser. Knots Everything*, pages 597–606. World Sci. Publ., Singapore, 2005.
- [28] Jesper Jacobsen and Paul Zinn-Justin. Algebraic Bethe Ansatz for the FPL^2 model. *J. Phys. A*, 37(29):7213–7225, 2004. [arXiv:math-ph/0402008](#).
- [27] Philippe Di Francesco, Paul Zinn-Justin, and Jean-Bernard Zuber. A bijection between classes of fully packed loops and plane partitions. *Electron. J. Combin.*, 11(1):Research Paper 64, 11 pp, 2004. [arXiv:math/0311220](#).
- [26] Paul Zinn-Justin. The asymmetric ABAB matrix model. *Europhys. Lett.*, 64(6):737–742, 2003. [arXiv:hep-th/0308132](#).
- [25] Paul Zinn-Justin and Jean-Bernard Zuber. Matrix integrals and the generation and counting of virtual tangles and links. *J. Knot Theory Ramifications*, 13(3):325–355, 2004. [arXiv:math-ph/0303049](#).
- [24] Gilles Schaeffer and Paul Zinn-Justin. On the asymptotic number of plane curves and alternating knots. *Experiment. Math.*, 13(4):483–493, 2004. [arXiv:math-ph/0304034](#).
- [23] Paul Zinn-Justin and Jean-Bernard Zuber. On some integrals over the $U(N)$ unitary group and their large N limit. *J. Phys. A*, 36(12):3173–3193, 2003. special issue on Random Matrix Theory. [arXiv:math-ph/0209019](#).
- [22] Jesper Jacobsen and Paul Zinn-Justin. Monochromatic path crossing exponents and graph connectivity in 2d percolation. *Phys. Rev. E*, 66(R):055102, 2002. [arXiv:cond-mat/0207063](#).
- [21] Paul Zinn-Justin. HCIZ integral and 2D Toda lattice hierarchy. *Nuclear Phys. B*, 634(3):417–432, 2002. [arXiv:math-ph/0202045](#).

- [20] Paul Zinn-Justin. The influence of boundary conditions in the six-vertex model. unpublished preprint, 2002. [arXiv:cond-mat/0205192](#).
- [19] Jesper Jacobsen and Paul Zinn-Justin. A transfer matrix for the backbone exponent of two-dimensional percolation. *J. Phys. A*, 35(9):2131–2144, 2002. [arXiv:cond-mat/0111374](#).
- [18] Jesper Jacobsen and Paul Zinn-Justin. Matrix models and the enumeration of alternating tangles. *Markov Process. Related Fields*, 9(2):301–310, 2003. proceedings of “Inhomogeneous random systems” (Cergy-Pontoise, 2002).
- [17] Jesper Jacobsen and Paul Zinn-Justin. The combinatorics of alternating tangles: from theory to computerized enumeration. In *Asymptotic combinatorics with application to mathematical physics (St. Petersburg, 2001)*, volume 77 of *NATO Sci. Ser. II Math. Phys. Chem.*, pages 97–112. Kluwer Acad. Publ., Dordrecht, 2002. [arXiv:math-ph/0111011](#).
- [16] Paul Zinn-Justin. The general $O(n)$ quartic matrix model and its application to counting tangles and links. *Comm. Math. Phys.*, 238(1-2):287–304, 2003. [arXiv:math-ph/0106005](#).
- [15] Jesper Jacobsen and Paul Zinn-Justin. A transfer matrix approach to the enumeration of colored links. *J. Knot Theory Ramifications*, 10(8):1233–1267, 2001. [arXiv:math-ph/0104009](#).
- [14] Jesper Jacobsen and Paul Zinn-Justin. A transfer matrix approach to the enumeration of knots. *J. Knot Theory Ramifications*, 11(5):739–758, 2002. [arXiv:math-ph/0102015](#).
- [13] Vladimir Korepin and Paul Zinn-Justin. Inhomogeneous six-vertex model with domain wall boundary conditions and Bethe Ansatz. *J. Math. Phys.*, 43(6):3261–3267, 2002. [arXiv:nlin.SI/0008030](#).
- [12] Paul Zinn-Justin. Six-vertex model with domain wall boundary conditions and one-matrix model. *Phys. Rev. E*, 62(3, part A):3411–3418, 2000. [arXiv:math-ph/0005008](#).
- [11] Vladimir Korepin and Paul Zinn-Justin. Thermodynamic limit of the six-vertex model with domain wall boundary conditions. *J. Phys. A*, 33(40):7053–7066, 2000. [arXiv:cond-mat/0004250](#).
- [10] Paul Zinn-Justin and Jean-Bernard Zuber. On the counting of colored tangles. *J. Knot Theory Ramifications*, 9(8):1127–1141, 2000. [arXiv:math-ph/0002020](#).
- [9] Paul Zinn-Justin. The six-vertex model on random lattices. *Europhys. Lett.*, 50(1):15–21, 2000. [arXiv:cond-mat/9909250](#).
- [8] Paul Zinn-Justin. Some matrix integrals related to knots and links. In *Random matrix models and their applications*, volume 40 of *Math. Sci. Res. Inst. Publ.*, pages 421–438. Cambridge Univ. Press, Cambridge, 2001. [arXiv:math-ph/9910010](#).
- [7] Paul Zinn-Justin and Jean-Bernard Zuber. Matrix integrals and the counting of tangles and links. *Discrete Math.*, 246(1-3):343–360, 2002. [arXiv:math-ph/9904019](#).
- [6] Paul Zinn-Justin. The dilute Potts model on random surfaces. *J. Statist. Phys.*, 98(1-2):j10–264, 2000. [arXiv:cond-mat/9903385](#).
- [5] Paul Zinn-Justin. Adding and multiplying random matrices: a generalization of Voiculescu’s formulas. *Phys. Rev. E* (3), 59(5, part A):4884–4888, 1999. [arXiv:math-ph/9810010](#).
- [4] Vladimir Kazakov and Paul Zinn-Justin. Two-matrix model with ABAB interaction. *Nuclear Phys. B*, 546(3):647–668, 1999. [arXiv:hep-th/9808043](#).
- [3] Paul Zinn-Justin. Nonlinear integral equations for complex affine Toda models associated with simply laced Lie algebras. *J. Phys. A*, 31(31):6747–6770, 1998. [arXiv:hep-th/9712222](#).
- [2] Paul Zinn-Justin and Natan Andrei. The generalized multi-channel Kondo model: thermodynamics and fusion equations. *Nuclear Phys. B*, 528(3):648–682, 1998. [arXiv:cond-mat/9801158](#).
- [1] Paul Zinn-Justin. Universality of correlation functions of Hermitian random matrices in an external field. *Comm. Math. Phys.*, 194(3):631–650, 1998. [arXiv:cond-mat/9705044](#).
- [0] Paul Zinn-Justin. Random Hermitian matrices in an external field. *Nuclear Phys. B*, 497(3):725–732, 1997. [arXiv:cond-mat/9703033](#).